Power BI Assignment 1

Data Transformation & Data Modeling – Solution Report

1. Data model view

Model view of the three tables, placed as saved in Power_BI_project_1.pbix. Column lists show the fields used in this assignment. Relationships follow the assignment specification.

Relationship	Key	Cardinality	Status
List of Orders → Order Details	Order ID	One-to-many (1:*)	Active
Order Details → Sales target	Category	Many-to-one (*:1)	Active

Modeling logic: Order ID is unique in List of Orders and repeats in Order Details (one order, many line items), so it is a clean 1:* link. Category is the shared field between line items and targets; it only works as a 1:* link if Sales target holds one row per Category. If it holds one row per Category per Month, Category repeats on both sides and Power BI will need Many-to-many, or a separate Category dimension table. Check this in Manage Relationships.

2. Transformation steps and logic

Task	What was done	Why
1. Import	Get Data → Text/CSV for List of Orders; New Source for Order Details and Sales Target, all in Power Query.	Keeps cleaning steps repeatable on refresh.
2. Types / rows	Keep Top Rows = 500; Order Date → Date (Using Locale, English (India) if needed); Amount and Target → Fixed decimal number.	Fixed decimal avoids floating-point rounding in currency.
3. Text	Customer Name → Transform → Format → Capitalize Each Word.	Consistent names for grouping and slicers.
4. New columns	Location = City & ", " & State via Merge Columns (comma). Profit Margin = [Profit] / [Amount], formatted as %.	Single geographic field; margin comparable across orders.
5. Conditional	Profit Status: Profit < 0 → Loss; = 0 → Break-Even; > 0 → Profit.	Quick profitable / unprofitable split.
6. Merge	Merge Queries on Order ID (List of Orders + Order Details), expand needed columns, rename to Orders Data.	Row-level detail alongside order header.
7. Missing / duplicates	Remove Duplicates on Order ID.	See note below.
8. Sort / filter	Orders Data sorted by Order Date descending; State filter set to Tamil Nadu for analysis.	Latest orders first; regional view.
9. Group By	Order Details copies: count rows by Order ID; average Profit by Category; sum Amount by Sub-Category. Sales Target copy: sum Target by month of order date.	Summary tables for quick KPIs.
10. Modeling	Relationships created in Model view as above, both active.	Filters flow from orders to details and targets.

Points worth checking before you submit

• **Remove Duplicates on Order ID:** Order Details normally has several rows per Order ID, so de-duplicating it on that column deletes valid line items and understates Amount and Profit. Apply it to List of Orders only, or use all columns on Order Details. The assignment also asks you to identify missing values (Table.Profile, or Column quality in View) and justify each choice: replace, remove or keep null. This step needs that write-up.

• **Merge column name:** the assignment text says "Location as State and City", but the required format is City, State and the name should be just `Location`.

• **Order of steps:** Keep Top 500 Rows before the merge, so Orders Data holds only those orders.